Dalyn Wachala

Mechatronics Engineering Student

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PROFESSIONAL SUMMARY

MTSU Mechatronics Engineering senior and aspiring controls engineer with hands-on experience spanning academic robotics research, competitive robotics, industrial process engineering, and technical teaching. Comfortable across the full stack of a robotics project, mechanical design, electronics, embedded programming, and simulation, with a particular interest in robotic systems and controls. Dedicated to pursuing advanced academic study to solve the next generation of complex mechatronic challenges.

EDUCATION

Middle Tennessee State University | 3.865 GPA

- B.S. in Mechatronics Engineering with a minor in Mathematics | August 2023 – May 2027
- Recipient of Presidential Scholarship from the school. Dean's List for 5 out of 6 semesters.

WORK EXPERIENCE

NSF Research Experience for Undergraduates (REU) – Robotics in Appalachia | May 2026 – July 2026

Interdisciplinary Robotics Laboratory (IRL), West Virginia University – Mentor: Dr. Yu Gu

- Researched terrain traversability estimation for field robots, developing a piecewise Gaussian Process Regression framework to map coefficient of friction (CoF) across terrain regions that cannot be directly sampled.
- Designed and built a custom 2-DOF friction probe and derived a kinematic/virtual-work-based model to compute static CoF from joint torque and drag measurements.
- Presented the project as a research poster to REU faculty, mentors, and peers as part of a 3-person research team.

Teaching Assistant | January 2026 – Present

Middle Tennessee State University

- Run weekly sessions for a class of 60 students, walking through tough problems in equilibrium and structural analysis to help them learn the material.
- Tutor students one-on-one and in groups, helping them work through difficult topics like truss analysis, friction, and internal forces.
- Grade assignments and help choose practice problems for the class, making sure the work stays at the high standard expected for a core engineering course.

Robotics Research Assistant | December 2025 – Present

Middle Tennessee State University

- Developing an Unmanned Aerial Vehicle (UAV) platform from the ground up, integrating high-performance motors, ESCs, and communication hardware for autonomous flight research.
- Configuring and programming Pixhawk flight controllers using Linux-based environments to implement custom flight parameters and advanced motor control algorithms.
- Executing precision hardware integration through professional soldering and electrical troubleshooting to ensure reliable system communication between transmitters and receivers.

Process Engineering Intern | January 2025 – January 2026

Valeo

- Optimized assembly line performance by diagnosing and repairing complex mechanical and electrical faults, improving throughput and minimizing downtime.
- Collaborated on the design and implementation of new assembly lines, assisting in technical setup and integration of part-specific production systems.

PROJECTS

3lb Combat Robot: Vertical Spinner • Micromouse Robot • Custom Speaker • Autonomous Turret System • Terrain Traversability Estimation via Gaussian Process Regression

Full write-ups, demos, and technical details available on my portfolio: dalynwachala.com

SKILLS

Software & OS: Linux (Ubuntu), ROS, Unity, Google Sheets, Arduino IDE, SolidWorks.

Electronics & Hardware: Soldering, Motor Control (ESCs), Pixhawk, Circuit Analysis, PCB Design (KiCAD), Sensor Design.

Robotics & Controls: Gaussian Process Regression, Traversability Estimation, Kinematics/Jacobian Analysis, Robot Simulation.

Manufacturing: 3D Printing (Klipper/Ender 3 v2), Manual Assembly, Simple Machining (Drill Press, Welding, Band Saw).

Soft Skills: Public Speaking, Technical Tutoring, Project Management, Research Presentation.

LEADERSHIP AND INVOLVEMENT

MTSU Robotics Club | Vice President

- Lead club operations and coordinate team projects, helping manage a group of students interested in hands-on robotics and competitive engineering.

Engineering Technology Department | Student Ambassador

- Serve as a representative for the department, giving tours and answering curriculum questions for prospective students and their families.